HookNet: Efficient Multiscale Context Aggregation for High-Accuracy Detection of Fabric Defects

Zhong Xiang, Yujia Shen, Miao Ma, and Miao Qian

***Abstract*— Automatic defect detection is a critical stage of quality control in the textile industry. Because defects vary in texture structure, size, and spatial distribution, existing defect detection methods have difficulty in achieving a perfect tradeoff between detection efficiency, accuracy, and generalizability. This article presents an advanced fabric defect detection model based on a deep learning algorithm. A novel parallel dilated attention module (PDAM) was designed: this is nested in the deep layers of the neural network backbone to establish global channel dependencies and capture multiscale contextual information. Furthermore, a hook-shaped feature pyramid network (HookFPN) was developed for multiscale context aggregation: this directs the network to focus on lower level features and is helpful for further improving detection efficiency without sacrificing detection accuracy. In addition, the Alpha-GIoU loss is used to improve the accuracy of bounding box regression because of its modulated power parameter α. The advantages and effectiveness of the proposed method were statistically analyzed using data from several public datasets. The code is available at https://github.com/aabb605/HookNet.**

***Index Terms*— Deep learning, defect detection, multiscale context aggregation, textile fabric.**

I. Introduction

IN THE process of textile production, distinct defects that vary in physical size and shape are randomly distributed across the fabric surface because of factors such as machine failure, damage, and contamination. Such defects have an adverse impact on both the quality and price of textile end products [1]. Because traditional manual inspection methods are dependent on personal experience and susceptible to fatigue and other drawbacks, machine-vision-based models and algorithms have been proposed in recent decades to automate the detection process. These can improve the accuracy or efficiency of defect detection to ensure the consistent quality of the end product [2]. However, it is still challenging for existing techniques to balance the detection efficiency, accuracy, and generalizability, and there are growing concerns from the textile industry and academia.

Technically, automatic methods for fabric defect detection can be categorized into three types: traditional methods, classical object detection models, and deep-learning-based methods. According to the image processing techniques used, existing traditional fabric defect detection methods are divided into four main categories: spectral methods, structural methods, statistical methods, and model-based methods [3]. For example, Hu and Wang [4] proposed a data fusion scheme based on undecimated wavelet decomposition to obtain a fused feature map. The distribution of pixel values corresponding to the defect-free background texture in the feature map is modeled according to the Gumbel distribution model. This method is effective for reducing the influence of uneven lighting and has good anti-interference properties. Ji et al. [5] proposed a weighted low-rank decomposition model with Laplace regularization to process images preprocessed by a Gabor filter, to increase the contrast between background regions and defect regions. Because it is difficult to dynamically determine the optimal filter parameters for each texture feature, the adaptability of this method is limited. Li et al. [6] proposed to combine a histogram of oriented gradients (HOG) with a low-rank decomposition model to improve the model's adaptability on complex textures and achieved good detection results. However, because of the very high computational cost of HOG operator inference, the detection speed of this method is limited. Di et al. [7] used quaternion images to represent fabric images and used a 2-D fractional Fourier transform to obtain the saliency map of the quaternion images. Their method has good robustness and performs well in the detection of different types of defects but cannot effectively detect defects that are randomly distributed in space. Although these traditional methods can achieve satisfactory results when applied to specific texture and pattern backgrounds, to comply with modern production requirements, advanced methods should be capable of handling complex and varied image texture patterns using suitable hardware equipment.

The convolutional neural network (CNN), which consists of convolutional and pooling layers [8], has excellent performance with respect to concurrent processing, feature extraction, and generalizability. With the increased availability of GPU computing power [9], object detection has become one of the most popular research areas in recent years, and many object detection models have been applied to vision tasks [10], [11]. The current mainstream neural network models can be divided into two categories: anchor-based detectors and anchor-free detectors. Anchor-based detectors can be further classified into two-stage detectors and one-stage detectors.

Manuscript received 22 January 2023; revised 11 April 2023; accepted 1 May 2023. Date of publication 5 June 2023; date of current version 14 June 2023. This work was supported by the Central Guidance on Local Science and Technology Development Fund of Zhejiang Province under Grant 2022ZYYDSA300214. The Associate Editor coordinating the review process was Dr. Jing Lei. *(Corresponding author: Zhong Xiang.)*

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Digital Object Identifier 10.1109/TIM.2023.3280519

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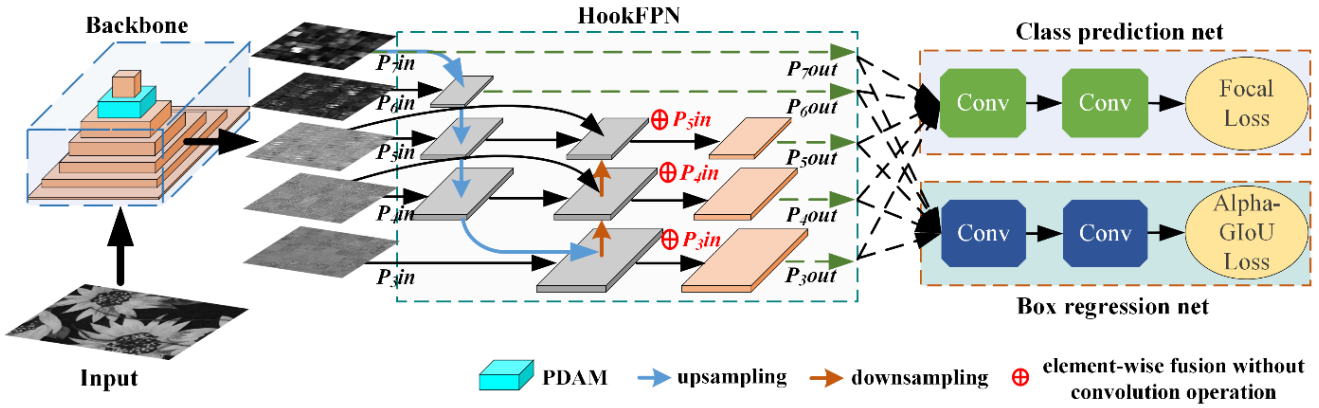

Fig. 1. Structure of the HookNet network framework.

Two-stage detectors (e.g., the R-CNN series [12], [13], [14], [15] and task-aware spatial disentanglement (TSD) [16]) first generate a series of candidate boxes as proposals and then classify and regress these proposals through a CNN. One-stage detectors (e.g., the YOLO series [17], [18], RetinaNet [19], single shot multibox detector (SSD) [20], and EfficientDet [21]) omit the process of generating proposals; they are designed to perform classification and regression tasks simultaneously. In recent years, anchor-free detectors (such as CornerNet [22], ExtremeNet [23], and fully convolutional one-stage object detector (FCOS) [24]) have been proposed, to eliminate anchor priors. For fabric defect detection tasks, anchor-based detectors perform better than anchor-free detectors because of the multiscale and random spatial distribution of the defect objects. Furthermore, to satisfy the speed requirements of real-time detection, one-stage detectors have a unique advantage in performance despite the higher detection accuracy of two-stage detectors [25]. However, current one-stage detectors are more suitable for natural scenes than textile images, and there are large differences in object properties (e.g., texture structure, size, and spatial distribution) between the two types of images. Therefore, the accuracy and speed of these existing deep CNNs (DCNNs) are still not adequate for fabric defect detection tasks.

Inspired by the successful improvement and application of DCNNs in the industry [26], [27], researchers have explored the potential of using deep-learning-based optimization methods to improve the performance of defect detection. Because of the diversity of textural structures and patterned backgrounds in fabrics, the adaptability of defect detection models has been a major concern for researchers. For example, Li et al. [28] used a wide-and-compact CNN structure for detecting defects in solid-color plain fabric. Xie and Wu [29] developed a RefineDet-based deep learning network and obtained good detection results on fabric with regular or irregular patterns. A link-SE module and dilation up-weight CAM (DUW-CAM) was proposed by Liu et al. [30] to improve the adaptability of a weakly supervised network to complex fabric textures. Zhang et al. [31] verified the performance of three variants of YOLOv2 models and proposed a localization and classification method for defects in yarn-dyed fabric. Detection accuracy is a key criterion for assessing whether a model can be applied in practice. Therefore, in the pursuit of good detection performance, Jing et al. [32] applied the K-means algorithm to determine the size of prior frames and added an additional detection layer to YOLOv3 to optimize the model structure. Jin and Niu [33] used a teacher–student architecture to improve the accuracy and efficiency of the YOLOv5 model in defect detection. To satisfy the speed requirements of real-time detection and to balance normal and abnormal samples, Jing et al. [34] proposed an extremely effective CNN, Mobile-Unet, for end-to-end defect segmentation. However, the above approaches cannot capture the rich contextual information of multiscale fabric defects efficiently because they ignore the relationship between the semantic nature of fabric defects and that of backgrounds while focusing on the improvement of detection performance metrics, that is, these methods fail to achieve a proper tradeoff between detection efficiency, accuracy, and generalizability.

To overcome the above drawbacks, a lightweight model for detecting fabric defects is proposed in this article. The detection model is based on EfficientDet-d0 and is named HookNet. Fig. 1 shows the overall framework of HookNet. A novel parallel dilated attention module (PDAM) was designed; this is nested in the deep layers of the neural network backbone to establish global channel dependencies and capture multiscale contextual information. Furthermore, a hook-shaped feature pyramid network (HookFPN) was developed for multiscale context aggregation: this directs the network to focus on lower level features and is helpful for further improving detection efficiency without sacrificing detection accuracy. In addition, the Alpha-GIoU loss is used to further optimize the regression effect of bounding boxes and enhance the ability of the model to localize the defect regions. The main contributions of this article are summarized as follows.

1) The PDAM, nested in the deep layers of the backbone, was designed. It is used to obtain a better interaction of information between local and global channels and thereby capture further contextual information in images of the fabric surface.
2) A biased feature fusion pyramid structure, named HookFPN, was designed to strengthen the weight of shallow feature information in the features extracted

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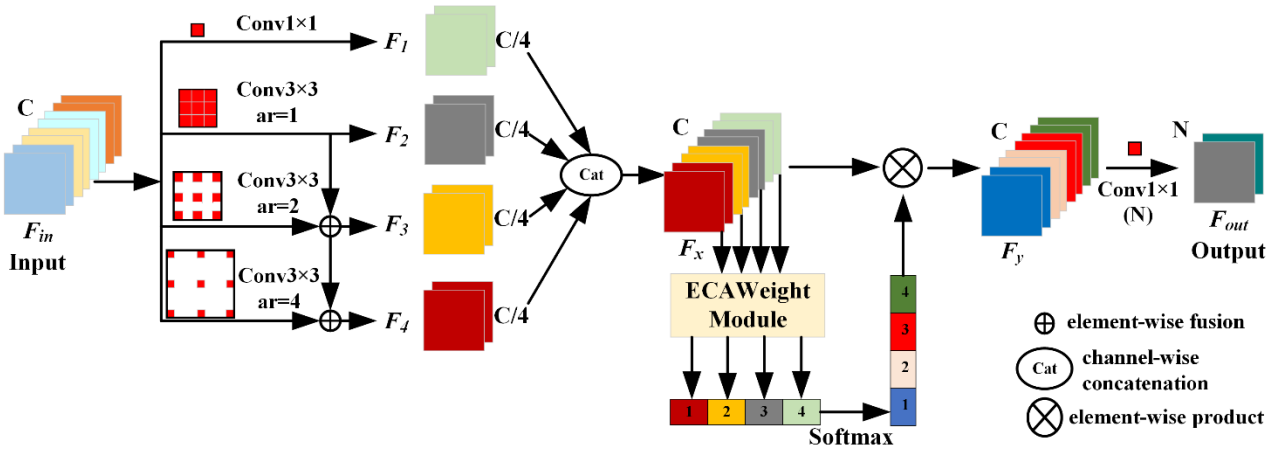

Fig. 2. Structure of the PDAM.

from images of fabric with complex texture and thereby improve the accuracy of defect detection.

3) The effect of Alpha-GIoU loss, the regression loss function based on intersection over union (IoU) loss and its variant, on the model performance was investigated and verified to provide an effective reference for researchers in the field of fabric defect detection.

II. Methods

A. Parallel Dilated Attention Module

The contextual information of visual objects characterizes the correlation between the semantic nature of the visual object and that of the visual scene [35]. Fabric images acquired in industrial production conditions often contain rich contextual information of multiscale fabric defects. In addition, in a fabric surface defect dataset, the size of defects varies greatly and the defects are randomly distributed at arbitrary locations in the images.

Inspired by the above idea, we propose the PDAM to capture the contextual information embedded in the fabric images; its structure is shown in Fig. 2. The refined output from the PDAM is obtained by the following steps. First, features at different scales are extracted from each branch by the hybrid dilated convolutional structure. Second, the channel-wise attention vectors of the multiscale feature maps are obtained by the efficient channel attention weight (ECAWeight) module to establish independent cross-channel interactions in a local manner. Finally, the whole multiscale channel attention vector is recalibrated by the Softmax function, which effectively establishes global channel dependencies. The dilated convolution has a large impact on the receptive field of the high-level feature maps. To fully exploit the performance of the PDAM, it is attached after the P5 layer of the backbone, to replace the original downsampling operation. The output of the PDAM acts as the input to the following P_7 layer of the backbone. The PDAM structure is explained in detail next.

The spatial information of the input feature is integrated by a hybrid dilated convolutional structure, which consists of one 1×1 convolutional branch and three dilated convolutional branches with various dilation rates. Cascade operations are implemented between convolutional branches with different dilation rates to alleviate the grid effect caused by a single convolutional branch; these operations help to obtain more information about global feature detail and enhance the inference capability of the network. Similarly, the 1×1 convolutional branch preserves as much detailed information as possible from the image without changing the feature map size. The subfeature maps $F_i (i = 1, 2, 3, 4)$ output by the hybrid dilated convolutional structure have the common channel dimension $C/4$, where C is the input channel dimension. All subfeature maps F_i are concatenated channel-wise to obtain the intermediate feature map F_x.

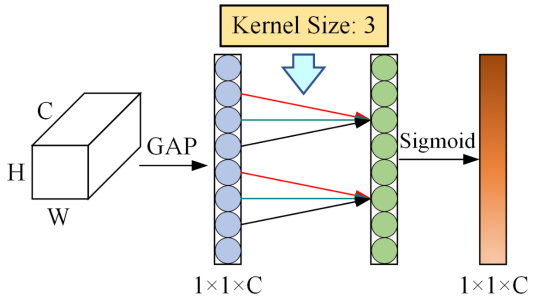

Fig. 3. Diagram of the ECAWeight module. Given the aggregated features obtained by GAP, it performs the 1-D convolution of size k followed by a sigmoid function to generate channel weights.

The ECAWeight module [36] obtains the importance of each feature channel in the intermediate feature map by assigning different weights to different channels. As shown in Fig. 3, after global average pooling (GAP) processing without dimensionality reduction, the ECAWeight module captures local cross-channel interaction by using the 1-D convolution. Note that kernel size k represents how many neighbors participate in the attention prediction of one channel. The corresponding weights of each channel are then adjusted to between 0 and 1 by a sigmoid function. In this manner, the attention weight vectors of different scales can be obtained from F_x. The partial vectors are defined by (1). The whole multiscale channel attention vector V ($V \in R^{c \times 1 \times 1}$) integrates the above local

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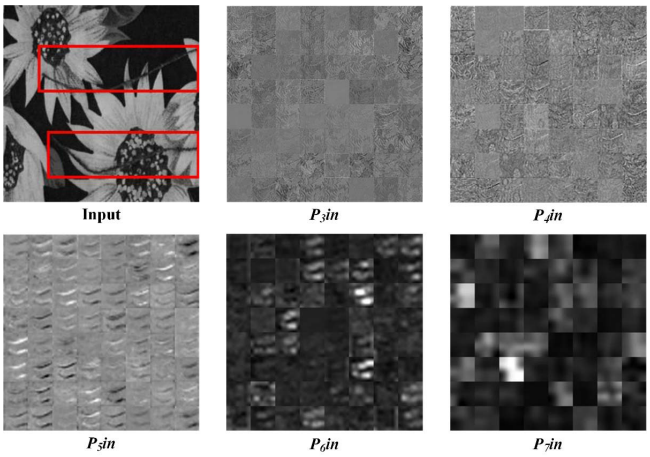

Fig. 4. Visualization of features with layers of different levels.

cross-dimensional vectors and is obtained by concatenation, as formalized by (2). To achieve interaction between local and global channel attention, the Softmax function is used across channels to adaptively select different spatial scales [37] and obtain the recalibrated weights W_i. The partial recalibrated weights W_i are defined by (3) and the whole recalibrated channel attention vector W is defined by (4)

$$V_i = \text{ECAWeight}(F_i)(i = 1, 2, 3, 4) \tag{1}$$

$$V = V_1 \oplus V_2 \oplus V_3 \oplus V_4 \tag{2}$$

$$W_i = \text{Soft}\max(V_i) = \frac{\exp(V_i)}{\sum_{i=1}^{S} \exp(V_i)}(i = 1, 2, 3, 4) \tag{3}$$

$$W = W_1 \oplus W_2 \oplus W_3 \oplus W_4. \tag{4}$$

Most importantly, the refined intermediate feature map F_y is obtained by concatenation: the multiscale feature maps F_i are multiplied by the corresponding partial recalibrated weights W_i and concatenated. The final 1×1 convolution operation is used to adjust the number of channels. This process can be formalized as follows:

$$Y_i = F_i \otimes W_i \tag{5}$$

$$F_y = \text{Concat}([Y_1, Y_2, Y_3, Y_4]) \tag{6}$$

$$F_{\text{out}} = \text{Conv}_{1\times1}(F_y) \tag{7}$$

where $\otimes$ represents the channel-wise multiplication, Y_i denotes the partial recalibrated feature map, F_y denotes another intermediate feature map, and F_{out} denotes the final refined output.

In summary, the proposed PDAM is able to obtain a better interaction of information between local and global channel attention by using the hybrid dilated convolutional structure and the Softmax function. Therefore, the PDAM can effectively aggregate global features and capture multiscale contextual information, thereby improving the flexibility of the network for detecting multiscale defects.

B. Hook-Shaped Feature Pyramid Network

For the defect detection task, the shallow features of the CNN contain rich information about texture structure. As the number of network layers increases, the image texture details become increasingly blurred because of the large receptive field. The result may be that the geometric details disappear completely, while the semantic information continues to increase. Fig. 4 shows an input image together with feature maps of various levels; each feature map contains 64 channels. The original BiFPN structure in EfficientDet, which implements bidirectional information flow and weighted feature fusion, allows different network layers to integrate information the same number of times for both low- and high-level features. However, for fabric defect detection, the input image differs significantly from images of natural scenes. Fabric images are rich in texture information with simple semantic information [38]. To ensure that the network focuses more on low-level features with limited computational resources, the FPN needs to integrate information in a biased manner.

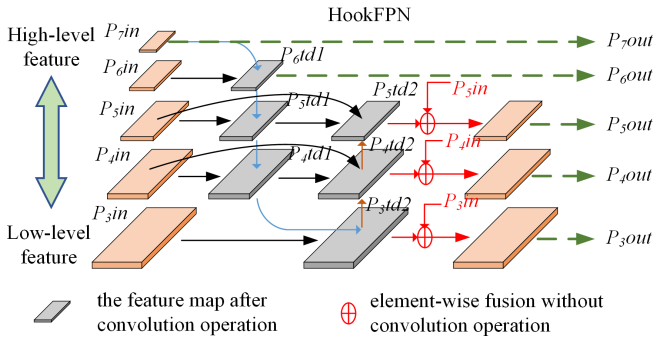

Fig. 5. HookFPN architecture design.

Inspired by the above ideas, we propose the HookFPN, whose structure is shown in Fig. 5. HookFPN extracts a specific feature P^{in} from the backbone and feeds the corresponding output feature P^{out} to the classification/regression network for subsequent operations. To characterize the output features of each layer in detail, the three intermediate feature maps (if they exist) at each level are defined as follows:

$$P_n td1 = \text{Conv}\left(\frac{\omega_1 \cdot P_n\text{in} + \omega_2 \cdot \text{Resize}(P_{n+1}td1)}{\omega_1 + \omega_2}\right) \tag{8}$$

$$P_n td2 = \text{Conv}\left(\frac{\omega_1' \cdot P_n\text{in} + \omega_2' \cdot P_n td1 + \omega_3' \cdot \text{Resize}(P_{n-1}td2)}{\omega_1' + \omega_2' + \omega_3'}\right) \tag{9}$$

$$P_n\text{out} = P_n td2 + P_n\text{in} \tag{10}$$

where $P_n td1$ is the first intermediate feature map at level n on the top-down pathway, $P_n td2$ is the second intermediate feature map at level n on the bottom-up pathway, $P_n$out is the final output feature at level n, and $\omega_i^{(p)}$ denotes the weights corresponding to the ith input edges of an intermediate node with a particular pathway. It should be noted that there are no intermediate features map $P_7 td1$ or $P_7 td2$ (generated at level 7) and the input P_7in is the final output at level 7, generated with no computational cost.

Compared with the classical BiFPN structure, HookFPN has three modifications. First, the computation times of feature

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fusion at high-level layers $\{P6, P7\}$ are appropriately reduced although the original bidirectional information flow is retained; this means that the high-level fusion nodes are selectively removed, that is, the HookFPN structure is designed to deliver information in a biased manner. Because the fusion nodes at the low level $\{P4, P5\}$ are not affected, the additional information aggregation branches from the input nodes to output nodes are retained. Second, to make full use of the initial features extracted by the backbone, HookFPN superimposes the intermediate feature P_ntd2 of the low level $\{P3, P4, P5\}$ with the corresponding inputs by summation and feeds the fused outputs to the classification/regression network for subsequent operations. This operation integrates rich features without adding significant computational cost. Its structure follows the design ideas described at the beginning of this section to perform feature fusion in a biased manner. Third, in contrast to EfficientDet, which treats BiFPN as one feature network layer and repeats the same layer multiple times, HookFPN removes subsequent redundant layers, depending on the complexity of the detection task. The single network layer of HookFPN reduces the model complexity, to some extent, and decreases the computational cost. In addition, it does not reduce the detection accuracy of the model. This simplification helps to satisfy the speed requirements of real-time detection.

Following information aggregation by HookFPN, the feature maps obtained from each effective level ($\{P3, P4, P5, P6, P7\}$) are transmitted independently to both classification/regression subnetworks for the next stages: defect classification and bounding box regression.

C. Loss Function

In an object detection task, the detector generally needs to implement both classification and regression. The one-stage detector generates thousands of anchor boxes, but the number of defect objects is very small for fabric surface images; therefore, the vast majority of anchor boxes are negative samples and there is a serious imbalance between the numbers of positive and negative samples. Therefore, the original focal loss [19] of the classification is retained in HookNet, and the loss is adjusted by setting a weight to adjust the loss contributions of positive and negative samples. In addition, the Alpha-GIoU loss [39] is used to replace the original Smooth L1 loss of the regression, to further improve the regression accuracy of the bounding box and provide the model with more flexibility by modulating α. This is because the process of judging the accuracy of positioning and retention of the predicted boxes B is based on the IoU metric, rather than the Euclidean distance between the four corners of the predicted boxes B and the ground-truth boxes B^{gt}. The total loss function L_{total}, classification loss L_{cls}, and regression loss L_{reg} of the training stage (mentioned above) are defined as follows:

$$L_{\mathrm{total}} = L_{\mathrm{cls}} + L_{\mathrm{reg}} \tag{11}$$

$$L_{\mathrm{cls}} = -\alpha_1(1-p_t)^{\gamma}\log(p_t) \tag{12}$$

$$L_{\mathrm{reg}} = 1 - \mathrm{IoU}^{\alpha_2} + \left(\frac{|C/(B \cup B^{\mathrm{gt}})|}{|C|}\right)^{\alpha_2} \tag{13}$$

where α_1 is used to balance the importance of the positive and negative examples, γ denotes a prefixed positive scale value, p_t denotes the posterior probability, α_2 is a single power parameter, and C denotes the smallest convex box enclosing B and B^{gt}.

GIoU [40] maximizes the area of the intersection between the predicted box and the ground-truth box while minimizing the area of the convex box. GIoU is designed to avoid the problem of gradient disappearance when the two boxes do not overlap. Alpha-GIoU uses the same power parameter α_2 for an IoU term and an additional penalty term, based on GIoU. By modulating the parameter α_2, Alpha-GIoU offers the flexibility to achieve different levels of bounding box regression accuracy. Therefore, Alpha-GIoU loss is used in HookNet to optimize model training and thereby improve the detection performance. In this study, α_1 was set to 0.25, γ was set to 2, and α_2 was set to 3.

III. Experiments

All experiments in this study were conducted on a computer with the Ubuntu Linux operating system. The processor was an Intel[1] Xeon[1] Silver 4214 CPU at 2.20 GHz with 32 GiB of RAM and the GPU was an NVIDIA GeForce RTX 2080 Ti. The Adam optimizer, with a weight decay of 4×10^{-5}, was used during the training process. The initial learning rate was set to 0.001 and then decayed. A total of 100 epochs were performed for the whole training process, and the initial batch size was set to 16.

A. Dataset and Metric

1) TILDA Dataset: TILDA is a publicly available and widely used fabric dataset. It contains eight representative textile fabrics (four unpatterned fabrics, two fabrics with regular patterns, and two fabrics with complex irregular patterns). The dataset contains images of fabric with defects in seven categories, based on the causes of the defects. To evaluate the detection performance of HookNet, four common fabric defect categories were used in this study, namely, wire, spot, hole, and dark thread. Fig. 6 shows each of these categories of defects, with two different pattern backgrounds or texture structures that belong to each category. The image size in the dataset is 768×512 pixels. In this study, the dataset was expanded by preprocessing using a series of common data enhancement methods (e.g., image flipping, rotation, cropping, brightness variation, contrast variation, and adding noise). Ultimately, 3198 images were used for training, 356 images were used for validation, and 790 images were used for testing.

2) DAGM2007 Dataset: DAGM2007, which is provided by the International Association for Pattern Recognition (IAPR) and the German chapter of the European Neural Network Society (GNSS), is a public fabric defect dataset. The dataset includes the labels of all defect images. There are 1000 normal images and 150 abnormal images for each category of defects, and the background of most images is a simple irregular pattern. Although the defect areas are generated

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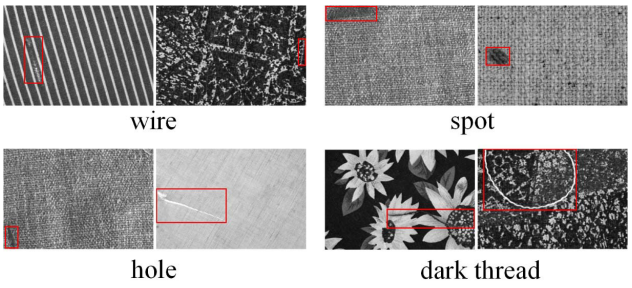

Fig. 6. Examples of defect images in the TILDA dataset.

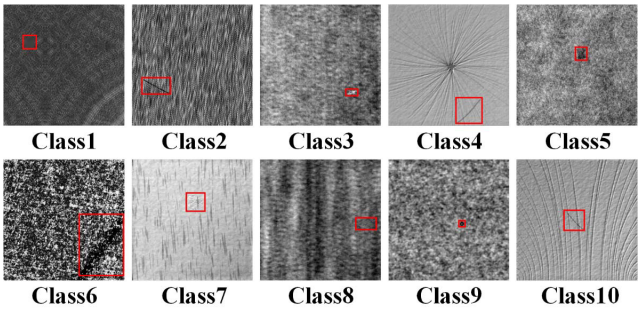

Fig. 7. Examples of defect images in the DAGM2007 dataset.

artificially, by various texture models, they are similar to actual defects. There are ten categories of defects, named Class 1, Class 2, . . . , Class 10, and the same background appears in all images in each category. Fig. 7 shows some typical defect images. The small difference in contrast between the defect itself and the background pattern makes manual annotation challenging. The image size is 512 × 512 pixels. After the same enhancement as that performed on TILDA images, there were 4320 images in the training set, 480 images in the validation set, and 1200 images in the test set.

3) Evaluation Metrics: Six common evaluation metrics for object detection were used in this study: precision, recall, F1, mean average precision (mAP), model parameters, and frame rate (FPS). Average precision (AP) is the area of the region enclosed between the precision−recall curve and the coordinate axis. The composite metric mAP is the mean of the AP values of the test results for all categories. Similarly, the F1 is calculated using a combination of precision and recall values. In addition, the model parameters and FPS metrics are used to quantify the complexity of the model. The above metrics are defined by the following formulas:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{14}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

$$F1 = \frac{2 \times P \times R}{P + R} \tag{16}$$

$$\mathrm{AP} = \int_0^1 P(R)dR \tag{17}$$

$$\mathrm{mAP} = \frac{1}{C}\sum_{k=1}^{C} \mathrm{AP}_k \tag{18}$$

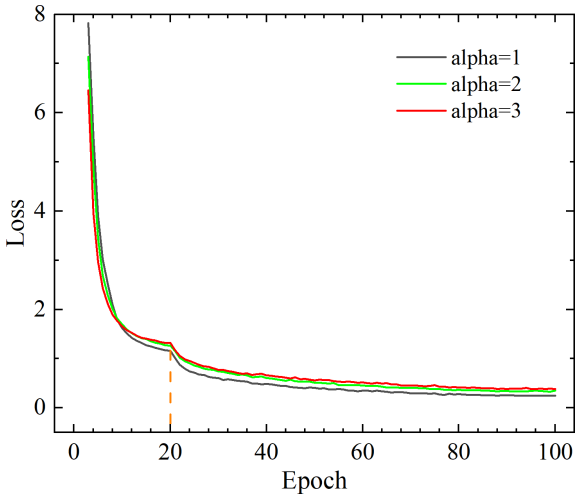

Fig. 8. Loss curve during training.

where C is the number of defect categories, P is the precision, R is the recall, and TP, FP, and FN are the numbers of true positives, false positives, and false negatives, respectively.

B. Analysis Based on TILDA Dataset

In these experiments, Alpha-GIoU loss, a flexibility IoU-based loss, was used to improve the accuracy of bounding box regression, and the pretrained model used for the experiments was publicly available (https://github.com/zylo117/Yet-Another-EfficientDet-Pytorch). In previous experiments [39], the parameter $\alpha = 3$ was found to perform better than $\alpha = 2$ in most cases. In the experiments reported in this section, modulation experiments were used to test the effects of different values of α during training and to determine the optimal value of the parameter for use in subsequent experiments. To improve the training efficiency, frozen training was used for the first 20 epochs: the backbone was frozen and the weights of the feature extraction network were not changed. Unfrozen training was then performed when the backbone was unfrozen and all parameters of the network were adjusted.

The model training loss values are shown in Fig. 8. In the first 20 epochs of the training stage, the loss values for the three values of the α parameter decreased considerably; after 60 epochs, all three loss curves converged. Fig. 9 shows the mAP curves of the validation set when the IoU threshold was set to 0.5. As the number of training epochs increased, the mAP values continued to increase until they became stable, and there was no overfitting. It is worth noting that, particularly around 40 epochs, the mAP value corresponding to $\alpha = 3$ was significantly greater than the mAP values corresponding to the other two values of α. Ultimately, this achieved approximately a 1% improvement in mAP and $\alpha = 3$ was therefore adopted for the subsequent experiments.

Table I shows the performance of HookNet and several publicly available object detection models (SSD [20], Faster RCNN [15], EfficientDet [21], YOLOv5, and improved RefineDet [29]). All the above methods were trained using pretrained models based on the TILDA dataset. As shown in Fig. 6, the texture of wire closely resembles its background

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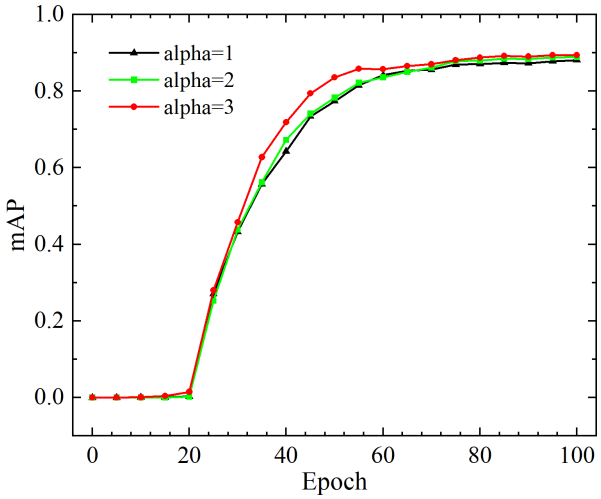

Fig. 9. mAP curve of the validation set.

pattern. In contrast to real-world images of natural scenes, the characteristics of the objects make it inevitable that fabric defect samples are manually annotated with complex texture and background information. These factors can interfere with the subsequent inspection process and make the task of detecting fabric defects challenging.

Compared with Faster RCNN and SSD, HookNet demonstrates a comprehensive advantage in terms of detection accuracy (precision, recall, F1, and mAP), model parameters, and detection speed. EfficientDet often fails to detect defect areas and therefore achieves the lowest recall. Although the improved RefineDet achieves the highest FPS and recall of the six methods, it is unstable for industrial use. One reason is that the accuracy greatly influences its reliability and does not allow for an effective balance between precision and recall; another reason is that the number of parameters (43.1M) does not satisfy the requirements for the deployment of a lightweight model in realistic environments. The FPS values measured for YOLOv5 and HookNet are similar, and HookNet has 2.7M fewer parameters than YOLOv5. Although YOLOv5 achieves a higher recall, its precision, F1, and mAP are all marginally lower than those of HookNet. Some of the defects are strongly similar in style to their backgrounds, and this leads to major disturbances in the detector. The proposed PDAM effectively captures multiscale contextual information and the proposed HookFPN focuses on lower level geometric details. Therefore, HookNet is marginally better than YOLOv5 in terms of detection accuracy (i.e., mAP). Although there is a significant difference between the FPS values measured for HookNet and those measured for the improved RefineDet, the FPS of HookNet still exceeds 30, and HookNet is, therefore, able to detect defects in real time. Experimental results on the TILDA dataset show that HookNet effectively captures the contextual information of objects and has a strong ability to encode features of hard samples, such as wire-like defects. In summary, HookNet has optimal detection performance with few parameters and has been proven to be highly suitable for fabric defect detection tasks in industrial scenarios.

Fig. 10 shows the detection results of several publicly available models on the TILDA dataset. The models frequently miss defect regions due to the multiscale characteristic of the defects and the interference of the background pattern. Moreover, the results also show that our proposed HookNet can better characterize the fabric texture and get good results.

TABLE I
COMPREHENSIVE COMPARISON ON THE TILDA DATASET

Method	P(%)	R(%)	F1(%)	mAP(%)	Parm.	FPS
SSD	78.83	72.04	75.28	74.78	24.2M	17.6
Faster RCNN	78.07	73.42	75.67	76.60	41.1M	12.7
EfficientDet-d0	97.26	70.82	81.96	87.15	3.8M	22.3
YOLOv5(s)	93.42	83.80	88.35	88.66	7.0M	30.3
Xie et al. [29]	79.29	92.30	85.30	87.18	43.1M	41.5
Ours	98.37	81.02	88.86	89.11	4.4M	31.3

C. Analysis Based on DAGM2007 Dataset

To visually demonstrate the advantages of HookNet in terms of detection performance, Fig. 11 shows the visualized images of some defect regions in the DAGM2007 dataset, which contains images of backgrounds with simple regular and irregular patterns. The first column shows the ground truth; the second to seventh columns show the detection results generated by SSD, Faster RCNN, EfficientDet, YOLOv5, Xie and Wu [29], and HookNet, respectively. It is worth noting that the area of each of the ten categories of defects is marked with a bounding box of a different color.

It can be observed from Fig. 11 that these models achieve good detection results on images of some specific categories of defects (e.g., Classes 1–3 and Class 10). For the other categories, the differences between the detection results of HookNet and other state-of-the-art methods are mainly reflected in two aspects of performance: the bounding box regression region and the network inference capability. For example, SSD and EfficientDet fail to recognize Class 8 fabric defects. The interference of the background patterns (e.g., Classes 4, 6, and 9) causes Faster RCNN to incorrectly treat normal regions as abnormal samples. Although YOLOv5 recognizes each category of defect, it could not accurately annotate the defect regions (e.g., Classes 6, 7, and 10). Furthermore, the confidence level of its inference is low. Compared with YOLOv5, the improved RefineDet and HookNet are able to detect defects with a higher confidence value, partly because of their higher detection performance. For Class 5 defects, the improved RefineDet is significantly disturbed by the background during model inference, causing it to additionally annotate a background region. Due to the aggregation of multiscale contextual information by the PDAM and HookFPN, HookNet is highly adaptable to defects at different scales and captures rich texture details of defects. Compared with other methods, HookNet achieves accurate region localization of the smallest bounding box, particularly for images of Classes 6 and 7 defects. In summary, HookNet

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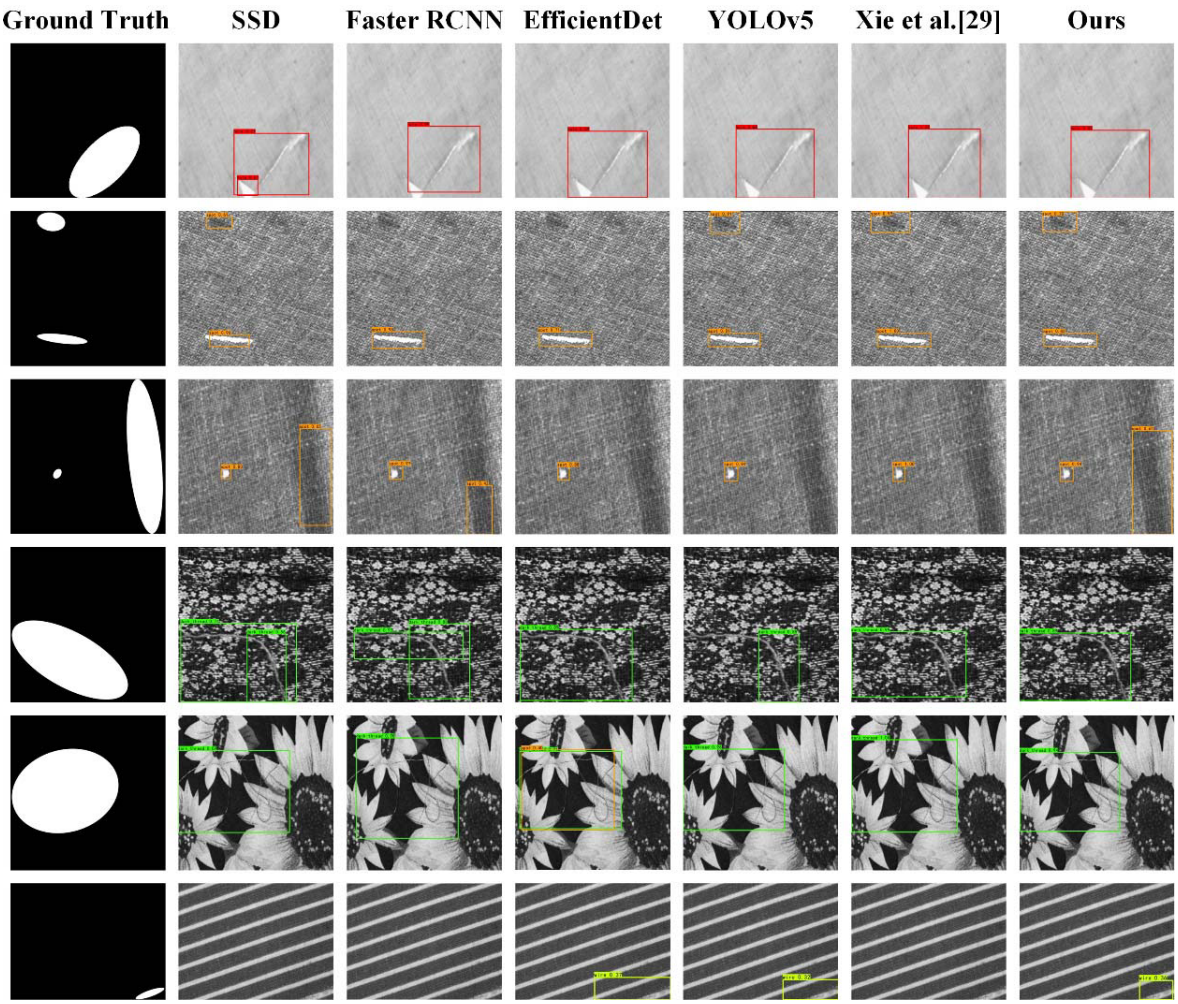

Fig. 10. Visual results of defect detection on the TILDA dataset.

TABLE II
COMPREHENSIVE COMPARISON ON THE DAGM2007 DATASET

Method	P(%)	R(%)	F1(%)	mAP(%)	Parm.	FPS
SSD	86.81	85.77	86.29	86.07	24.2M	27.8
Faster RCNN	87.56	89.69	88.61	88.60	41.1M	13.7
EfficientDet-d0	99.47	91.67	95.41	96.35	3.8M	24.4
YOLOv5(s)	97.42	95.56	96.48	97.33	7.0M	31.3
Xie et al. [29]	92.60	97.92	95.19	97.10	43.1M	41.7
Zhang et al. [41] (with K-means)	98.8	99.1	99.0	99.5	13.6M	-
Zhang et al. [41] (without K-means)	96.4	97.4	96.9	98.5	13.6M	-
Zhu et al. [42]	-	-	98.41 ±0.26	98.17 ±0.28	-	19.2
Ours	99.44	93.83	96.55	98.56	4.4M	33.5

is extremely suitable for fabric defect detection and achieves the best detection results.

To further demonstrate the superior performance of HookNet, Table II shows the results of HookNet and other advanced detectors on the DAGM2007 dataset. The structural properties of the defects are very similar to those of the corresponding backgrounds, and the scale difference between the defects is significant; these characteristics add to the challenge of the detection task. SSD is a classical one-stage detector, whose detection speed is significantly higher than that of Faster RCNN, a two-stage detector. From Table II, it can be observed that neither of these models can be applied directly to vision tasks in industrial scenarios because of their limited detection performance. The advantage of EfficientDet, a lightweight object detection model, is its small number of parameters (3.8M). However, the model is inferior to YOLOv5, improved RefineDet, and HookNet in terms of detection accuracy and detection speed. YOLOv5 is one of the most dominant object detection models and it excels in a variety of visual scene detection tasks. Compared with YOLOv5, HookNet achieves a 2.2 improvement in FPS, approximately 2.6M reduction in the number of model parameters, 1.23% improvement in mAP, 0.07 improvement in F1, and 2.02% improvement in precision. Although the improved RefineDet achieves satisfactory performance in terms of detection speed and recall, its overall performance is not satisfactory because of its low precision. Zhang et al. [41] proposed a novel network for surface defect detection. The proposed network

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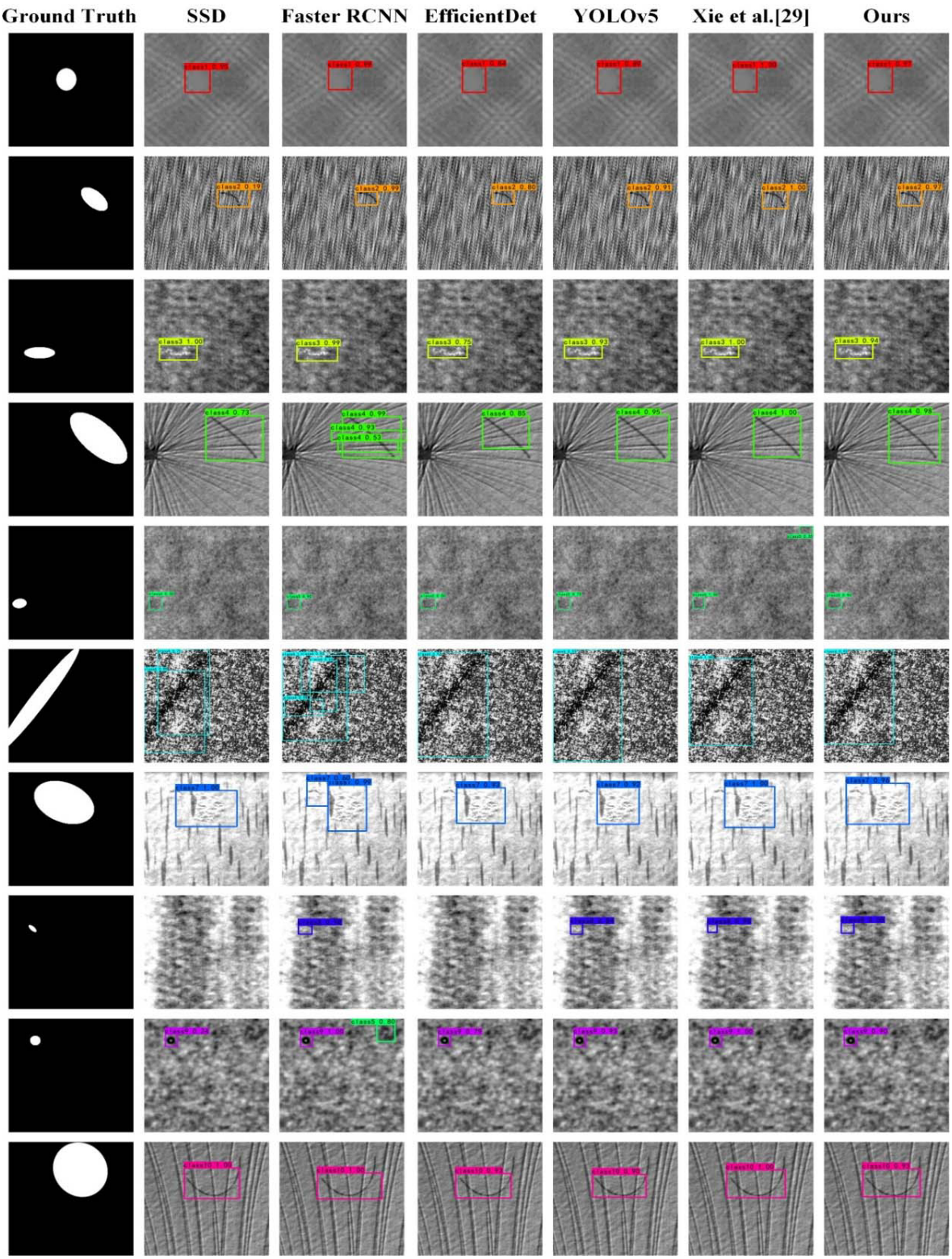

Fig. 11. Visual results of defect detection on the DAGM2007 dataset.

achieves the mAP of 98.5% and has a recall of 97.4% while maintaining a precision of 96.4%, which is roughly the same performance as our proposed HookNet. Based on such performance, the network improves the mAP by 1.0% due to the use of an additional optimization strategy (K-means clustering for anchors adjustment) in the preprocessing stage. HMFCA-Net [42] achieves the mAP of 98.17% and FPS of 19.2 on the DAGM2007 dataset. In fact, the structure of the model shows that it has more than 30M parameters. Compared to our HookNet, the model has a significant limitation in terms of the number of parameters and detection speed. From the experimental results, it can be seen that the precision and recall of all the above models are often mutually constrained, and HookNet achieves a better tradeoff between detection speed

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TABLE III
PERFORMANCE COMPARISON OF THE COMPONENTS ON THE TILDA DATASET

Group	Method	mAP(%)	GFLOPs	FPS
1	EfficientDet-d0 (baseline)	87.15	4.76	22.3
2	EfficientDet-d0 + HookFPN	87.34	4.61	32.2
3	EfficientDet-d0 + Alpha-GIoU	88.20	4.76	22.6
4	EfficientDet-d0 + PDAM	88.75	4.85	21.1
5	EfficientDet-d0 + Alpha-GIoU + PDAM	89.02	4.85	21.2
6	EfficientDet-d0 + HookFPN + Alpha-GIoU + PDAM (HookNet)	89.11	4.70	31.3

and accuracy than other state-of-the-art methods. In summary, HookNet is applicable to the task of detecting defects on both unpatterned fabrics and simple irregularly patterned fabrics. It also has the potential to improve the performance if some additional optimization strategies (e.g., K-means clustering for anchors adjustment or non-maximum suppression (NMS) variants during postprocess) are implemented. Furthermore, since the model has only 4.4M parameters, it satisfies the requirements for lightweight mobile deployment.

D. Ablation Experiments

To comprehensively evaluate HookNet and to obtain a clear role of each module on the overall performance, ablation experiments are conducted as described in this section. These experiments strip out parts of the components so that the selected parts could be evaluated, with respect to both speed and accuracy, to determine their impact on the overall effectiveness of the model. As shown in Table III, HookNet comprises four parts: EfficientDet-d0, HookFPN, Alpha-GIoU loss, and PDAM. Finally, mAP, GFLOPs, and FPS are chosen as the evaluation metrics. Note that the GFLOPs refer to the number of floating-point operations, which can be interpreted as the amount of computation. It is used to measure the complexity of the model and is often used as an indirect measure of inference speed.

A comparison of the experimental results of groups 1 and 2 shows a slight improvement in mAP (by 0.19%) and a significant increase in FPS (about 10) with HookFPN. Due to the simplified processing of the redundant feature network layer, the HookFPN structure is able to significantly increase the detection speed without reducing the detection accuracy, to satisfy the speed requirements of real-time fabric defect detection. Similarly, the results of groups 5 and 6 also validate the advantages of HookFPN. The results of groups 4 and 5 demonstrate that the use of Alpha-GIoU loss instead of Smooth L1 loss in the training stage slightly improves the detection performance of the model. Because there are no additional parameters with computational cost and there is no significant difference between the FPS values of the two groups. A comparison of the results of groups 1 and 4 shows that the proposed PDAM results in a 1.60% improvement in mAP. However, it introduces additional model parameters with computational cost, causing the FPS value to drop slightly (by 1.2). Ultimately, HookNet integrates the above four parts. Comparing the results of group 6 with the results of other groups, it can be observed that the composite model achieved the highest mAP; its high localization accuracy is shown in Fig. 11. Because of the additional computational cost introduced, the FPS value achieved by the composite model was less than that achieved by the single HookFPN component (group 2), but it still represents a substantial speed improvement over group 5. In summary, the experimental results demonstrate that HookNet achieves a reliable tradeoff between detection speed and accuracy.

IV. CONCLUSION

A lightweight method for fabric defect detection, HookNet, is proposed in this article. It includes the PDAM feature aggregation module and HookFPN network structure, which were specifically designed for the characteristics of fabric defect detection, and has demonstrated good detection performance. The PDAM establishes cross-channel interactions in a local and global manner to capture more contextual information. The HookFPN focuses on low-level features, reducing the model complexity. Both of these components are essential for fabric defect detection. In addition, Alpha-GIoU loss is used to further improve the localization accuracy of the bounding box. The experimental results show that HookNet is superior to existing methods for fabric defect detection tasks in industrial conditions. The main factor that hinders the accuracy improvement of HookNet is its relatively low recall, which is caused by the model missing defect regions during the detection process. Therefore, in future work, we will focus on increasing the recall of the model to minimize the number of omissions during the detection process as much as possible.

REFERENCES

[1] L. Shao, E. Zhang, Q. Ma, and M. Li, "Pixel-wise semisupervised fabric defect detection method combined with multitask mean teacher," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.

[2] H. Uzen, M. Turkoglu, and D. Hanbay, "Texture defect classification with multiple pooling and filter ensemble based on deep neural network," *Expert Syst. Appl.*, vol. 175, Aug. 2021, Art. no. 114838.

[3] C. Li, J. Li, Y. Li, L. He, X. Fu, and J. Chen, "Fabric defect detection in textile manufacturing: A survey of the state of the art," *Secur. Commun. Netw.*, vol. 2021, May 2021, Art. no. 9948808.

[4] G. H. Hu and Q. H. Wang, "Fabric defect detection via un-decimated wavelet decomposition and Gumbel distribution model," *J. Eng. Fibers Fabrics*, vol. 13, no. 1, pp. 15–32, 2018.

[5] X. Ji et al., "Fabric defect fetection via weighted low-rank decomposition and Laplacian regularization," *J. Eng. Fibers Fabrics*, vol. 15, pp. 1–14, Sep. 2020.

[6] C. Li, G. Gao, Z. Liu, D. Huang, and J. Xi, "Defect detection for patterned fabric images based on GHOG and low-rank decomposition," *IEEE Access*, vol. 7, pp. 83962–83973, 2019.

[7] L. Di, H. Long, and J. Liang, "Fabric defect detection based on illumination correction and visual salient features," *Sensors*, vol. 20, no. 18, p. 5147, Sep. 2020.

[8] S. H. Lee, C. S. Chan, S. J. Mayo, and P. Remagnino, "How deep learning extracts and learns leaf features for plant classification," *Pattern Recognit.*, vol. 71, pp. 1–13, Nov. 2017.

[9] K. Raju and N. N. Chiplunkar, "A survey on techniques for cooperative CPU-GPU computing," *Sustain. Comput.-Informat. Syst.*, vol. 19, pp. 72–85, Sep. 2018.

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[10] H. Fang, M. Xia, H. Liu, Y. Chang, L. Wang, and X. Liu, "Automatic zipper tape defect detection using two-stage multi-scale convolutional networks," *Neurocomputing*, vol. 422, pp. 34–50, Jan. 2021.
[11] Y. Zhao et al., "Simultaneous detection of defects in electrical connectors based on improved convolutional neural network," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.
[12] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.
[13] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.
[14] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[15] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[16] G. Song, Y. Liu, and X. Wang, "Revisiting the sibling head in object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11560–11569.
[17] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6517–6525.
[18] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
[19] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.
[20] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 21–37.
[21] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.
[22] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 765–781.
[23] X. Zhou, J. Zhuo, and P. Krähenbühl, "Bottom-up object detection by grouping extreme and center points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 850–859.
[24] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.
[25] M. Carranza-García, J. Torres-Mateo, P. Lara-Benítez, and J. García-Gutiérrez, "On the performance of one-stage and two-stage object detectors in autonomous vehicles using camera data," *Remote Sens.*, vol. 13, no. 1, p. 89, Dec. 2020.
[26] R. Miao et al., "Real-time defect identification of narrow overlap welds and application based on convolutional neural networks," *J. Manuf. Syst.*, vol. 62, pp. 800–810, Jan. 2022.
[27] A.-A. Tulbure, A.-A. Tulbure, and E.-H. Dulf, "A review on modern defect detection models using DCNNs–deep convolutional neural networks," *J. Adv. Res.*, vol. 35, pp. 33–48, Jan. 2022.
[28] Y. Li, D. Zhang, and D.-J. Lee, "Automatic fabric defect detection with a wide-and-compact network," *Neurocomputing*, vol. 329, pp. 329–338, Feb. 2019.
[29] H. Xie and Z. Wu, "A robust fabric defect detection method based on improved RefineDet," *Sensors*, vol. 20, no. 15, p. 4260, Jul. 2020.
[30] Z. Liu, Z. Huo, C. Li, Y. Dong, and B. Li, "DLSE-Net: A robust weakly supervised network for fabric defect detection," *Displays*, vol. 68, Jul. 2021, Art. no. 102008.
[31] H. Zhang, L. Zhang, P. Li, and D. Gu, "Yarn-dyed fabric defect detection with YOLOV2 based on deep convolution neural networks," in *Proc. IEEE 7th Data Driven Control Learn. Syst. Conf. (DDCLS)*, 2018, pp. 170–174.
[32] J. Jing, D. Zhuo, H. Zhang, Y. Liang, and M. Zheng, "Fabric defect detection using the improved YOLOv3 model," *J. Eng. Fibers Fabrics*, vol. 15, Mar. 2020, Art. no. 155892502090826.
[33] R. Jin and Q. Niu, "Automatic fabric defect detection based on an improved YOLOv5," *Math. Problems Eng.*, vol. 2021, Sep. 2021, Art. no. 7321394.
[34] J. Jing, Z. Wang, M. Rätsch, and H. Zhang, "Mobile-Unet: An efficient convolutional neural network for fabric defect detection," *Textile Res. J.*, vol. 92, nos. 1–2, pp. 30–42, Jan. 2022.
[35] V. Chalavadi, P. Jeripothula, R. Datla, S. C. Babu, and K. C. Mohan, "mSODANet: A network for multi-scale object detection in aerial images using hierarchical dilated convolutions," *Pattern Recognit.*, vol. 126, Jun. 2022, Art. no. 108548.
[36] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," 2019, *arXiv:1910.03151*.
[37] H. Zhang, K. Zu, J. Lu, Y. Zou, and D. Meng, "EPSANet: An efficient pyramid squeeze attention block on convolutional neural network," 2021, *arXiv:2105.14447*.
[38] T. Zhou, J. Zhang, H. Su, W. Zou, and B. Zhang, "EDDs: A series of efficient defect detectors for fabric quality inspection," *Measurement*, vol. 172, Feb. 2021, Art. no. 108885.
[39] J. He, S. Erfani, X. Ma, J. Bailey, Y. Chi, and X.-S. Hua, "Alpha-IoU: A family of power intersection over union losses for bounding box regression," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 20230–20242.
[40] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666.
[41] Y. Zhang et al., "Development of a cross-scale weighted feature fusion network for hot-rolled steel surface defect detection," *Eng. Appl. Artif. Intell.*, vol. 117, Jan. 2023, Art. no. 105628.
[42] Y. Zhu, R. Ding, W. Huang, P. Wei, G. Yang, and Y. Wang, "HMFCA-Net: Hierarchical multi-frequency based channel attention net for mobile phone surface defect detection," *Pattern Recognit. Lett.*, vol. 153, pp. 118–125, Jan. 2022.

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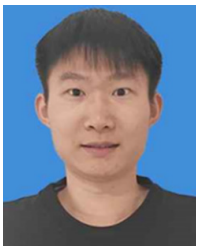

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